COSMIC BALANCE THEORY

One currency. One sum. Zero.

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Abstract

The Cosmic Balance Theory (CBT) rests on two axioms and a bridge: Mass is energy — there is only one currency. In a closed system, the sum of all forms of energy is conserved — there is a total. This total is zero — the cosmos is a zero-sum game. One currency, one sum, zero.

Three consequences follow from these axioms: Dark matter is collapsed energy (black holes). The baryon asymmetry is an observational artifact (B = 0). Dark energy is the geometric consequence of the zero sum. The CBT thereby addresses the DM problem, the baryon asymmetry problem, the DE problem, the coincidence problem, the flatness problem, and the Sakharov conditions — and circumvents the question “Why is there something rather than nothing?”

The CBT makes falsifiable predictions. The specific mechanisms that implement the consequences are not part of the core CBT — they are open tasks for theoretical physics. The individual building blocks exist in the literature (Tryon, Guth, Carr, García-Bellido, Dolgov). The combination B = 0 + PBH-DM + zero-sum DE has no published predecessor.

1. Axiom — Bridge — Axiom

Axiom 1: Mass Is Energy

There are no fundamentally different kinds of matter. There is only energy in different states. What we cannot measure with electromagnetic radiation, we call “dark.” Black holes are energy in its densest, non-radiating state. Visible matter is energy in a detectable state. Radiation is energy in motion. The 5/27/68 partition of the standard model is a classification by our measurement capability — not by physics.

Bridge: Conservation of Energy

Because there is only one currency — energy — a total sum can be formed. In a closed system, the sum of all forms of energy is conserved. It remains constant. The question is only: How large is this sum?

Axiom 2: +1 − 1 = 0

CBT postulates: This sum is zero. The cosmos is a zero-sum game. Positive energy (mass, radiation) is exactly compensated by negative energy (gravity). The zero-sum principle extends to all conserved quantum numbers: Not only is the total energy zero, but also the net baryon number, the net lepton number, and the total charge. What the standard model calls baryon asymmetry is a wrongly posed question.

> *One currency. One sum. Zero.*

2. The Three Consequences

Consequence 1: Dark Matter Is Collapsed Energy

Black holes are gravitationally blind to their content. A black hole formed from matter is physically identical to one formed from antimatter or pure radiation. Mass, spin, charge — that is all physics knows from the outside (no-hair theorem: Israel 1967, Carter 1971, Hawking 1972).

CBT identifies dark matter as primordial black holes (PBHs) — collapsed energy from the early universe. In the mass range 10^{14} to 10^{19} kg (asteroid to lunar mass), this is the largely open observational window in which PBHs can account for all of dark matter. Confirmed by Green (2024), Gorton & Green (2024), Carr et al. (2021), Green & Kavanagh (2021).

PBHs form during the radiation-dominated era from overdense regions. At that time, such a region contains everything — quarks, antiquarks, gluons, photons, leptons — at 106.75 relativistic degrees of freedom. The net baryon number accounts for less than 10^{-11} of the total energy. A PBH is more than 99.9999999% formed from symmetric energy.

Consequence 2: Antimatter Has Collapsed, Not Disappeared

The apparent baryon asymmetry ($\eta \approx 6 \times 10^{-10}$) is not a fundamental puzzle. Before PBH formation, the net baryon number was zero — equal amounts of matter and antimatter. Regions with a slight antimatter excess were statistically correlated with overdensities and preferentially collapsed into PBHs. After collapse, the baryon number is lost (no-hair theorem), but the corresponding positive baryon number remains as free matter.

The ALPHA-g experiment at CERN (Nature, 2023) confirms: Antimatter falls like matter. Gravity does not distinguish. What distinguishes is the state: collapsed or free.

The ratio $\Omega_{DM}/\Omega_b = 5.4$ is not a contradiction. Ω_{DM} is determined by the amount of energy that collapses into PBHs. Ω_b is determined by the amount of net baryon number that survives as free matter. Both depend on different physics.

Consequence 3: Dark Energy Is the Zero Sum Itself

In a closed zero-sum system: positive energy (mass + radiation) + negative energy (gravity) = 0. “Dark energy” is not a third substance alongside matter and radiation. It is the geometric consequence of the universe being flat ($\Omega_{total} = 1$, confirmed by Planck) — *because* it is a zero-sum game.

The zero-energy universe hypothesis is not new. Tryon (1973), Guth (1981), and Hawking argued that gravitational binding energy exactly compensates positive mass-energy. CBT elevates this hypothesis to an axiom and draws the consequence: The 68% “dark energy” is not “something unknown” — it is the difference that brings the sum to zero.

Standard Model	CBT Interpretation	Status
Visible matter (5%)	Free energy in detectable state	Observed
Dark matter (27%)	Collapsed energy (PBHs)	Falsifiable
Dark energy (68%)	Geometric zero-sum consequence	Consistent with $\Omega = 1$
Baryon asymmetry	Observational artifact ($B = 0$)	Falsifiable
Total: Three entities	**One process: Energy = 0**	

3. Falsifiable Predictions

A theory is scientific only if it is falsifiable. CBT makes the following specific predictions that distinguish it from ΛCDM:

Prediction	Method	Timeline
PBHs at 10^{14}–10^{19} kg as all of DM	Microlensing (Roman, Subaru), GW (Einstein Telescope), LISA (SIGW)	5–15 years
Sub-solar black holes without star-forming region	Gravitational waves (LIGO/Virgo/KAGRA, Einstein Telescope)	Now — S251112cm (Nov 2025) is consistent
Net baryon number B = 0	CMB isocurvature modes (LiteBIRD, CMB-S4)	~2032–2035
Stochastic GW background at ~mHz (from inflationary peak)	LISA	~2035
Ω_DM/Ω_b = 5,4 follows from one mechanism	Theoretical calculation (task for the community)	Open

Falsification criteria: If by ~2040 no PBH signature in the asteroid-mass range is found (via Roman, Einstein Telescope, LISA), Consequence 1 is refuted. If CMB-S4 finds large isocurvature modes incompatible with B = 0, Consequence 2 is refuted.

What CBT does *not* predict: A specific formation mechanism (QCD transition, inflationary peak, or other). A specific separation mechanism (Affleck-Dine or other). These are calculable implementations of the axioms — open tasks, not components of the theory.

4. Which Problems of Physics Does the CBT Address?

With two axioms, the CBT touches an unusually broad range of open questions. The following overview distinguishes between problems that are directly solved, those that are circumvented or reinterpreted, and those that are indirectly touched.

Directly Addressed

The dark matter problem. What is DM? CBT: Collapsed energy — black holes, primarily primordial. No new particles beyond the standard model required. The asteroid-mass window (10^{14}–10^{19} kg) is as of 2026 the largely open window for f_PBH = 1. After over 30 years of null results in direct WIMP searches (LZ, 2025: no signal in 417 live days), non-particle DM candidates gain plausibility.

The baryon asymmetry problem. Why is there more matter than antimatter? CBT: There is not. B = 0. Antimatter has collapsed, not disappeared. Regions with antimatter excess were statistically correlated with overdensities and collapsed into PBHs. The apparent asymmetry is an observational artifact. This eliminates the problem rather than solving it.

The dark energy problem. What is DE? CBT: Not a substance, but the geometric consequence of the zero sum. In a flat universe (Ω = 1), negative gravitational energy exactly compensates positive mass-energy. The 68% are the difference that brings the sum to zero.

The coincidence problem. Why are Ω_DM and Ω_b of similar order of magnitude (~5:1)? In most models, this is a coincidence. In CBT, both depend on the same fields: The inflationary peak determines Ω_DM, the baryon number correlation determines Ω_b. The ratio 5.4 becomes a prediction, not a coincidence.

Circumvented or Reinterpreted

The Sakharov conditions. Since 1967, three conditions have been considered mandatory for baryogenesis: B violation, C/CP violation, departure from thermal equilibrium. CBT circumvents all three: If B = 0, there is no baryon number to generate. The Sakharov conditions become irrelevant because the question was wrongly posed.

The flatness problem. Why is Ω = 1.000 ± 0.002? Standard inflation explains this dynamically. CBT goes further: The universe *must* be flat because total energy = 0 is consistent with a flat universe. Flatness is not a dynamical result but a logical requirement of the zero-sum axiom.

“Why is there something rather than nothing?” CBT: There is nothing. The sum is zero. “Something” is a local splitting into +1 and −1 of a total system that is exactly zero.

Indirectly Touched

Overmassive early black holes (JWST). JWST finds BHs of 10^7–10^8 M⊙ at z > 7 — too heavy for standard accretion. PBH seeding solves this: Massive PBHs with broad mass distributions serve as seeds for early SMBH growth.

The lithium problem of BBN. The predicted primordial ^{7}Li abundance is ~3× higher than observed. If a significant mass fraction goes into PBHs before BBN, this changes the local effective baryon density. Whether this solves the Li problem is an open quantitative question.

Sub-solar BH mergers (LIGO S251112cm). In November 2025, LVK reported a merger candidate with at least one component below 1 M⊙ (>99% probability). Stellar physics cannot produce BHs below ~2–3 M⊙. PBHs can. If confirmed, this would be the first direct hint of a primordial BH population.

Problem	Status	CBT Approach
What is dark matter?	**Directly solved**	Collapsed energy (PBHs)
Baryon asymmetry	**Directly solved**	B = 0, antimatter collapsed
What is dark energy?	**Reinterpreted**	Geometric zero sum
Koinzidenz Ω_DM/Ω_b ≈ 5	**Addressed**	Common origin
Sakharov conditions	**Circumvented**	B = 0 → no baryogenesis needed
Flatness problem	**Reinterpreted**	Flatness = logically required
Why something rather than nothing?	**Answered**	There is nothing. Sum = 0
JWST overmassive early BHs	Indirect	PBH seeding
LIGO sub-solar mergers	Consistent	PBH population expected

5. Predecessors and Demarcation

The CBT combines ideas that individually exist in the literature. No published model combines all three elements: DM = PBHs, B = 0, DE = zero sum. The following overview places the CBT within the existing research landscape.

PBHs as Dark Matter

The hypothesis that PBHs constitute dark matter dates back to Hawking (1971) and Chapline (1975). The modern formulation with detailed exclusion bounds comes from Carr, Green, Kavanagh, and others (2021). The asteroid-mass window was opened by Katz et al. (2018) and Smyth & Profumo (2020). CBT uses this window but does not invent it.

Common Origin of DM and Baryogenesis

García-Bellido et al. (2020, Phil. Trans. Royal Society A) developed a scenario in which PBHs at the QCD transition generate the baryon asymmetry through electroweak sphaleron transitions. Dolgov, Naselsky et al. (2000) discussed baryogenesis via PBH evaporation. Hook (2014) showed that asymmetric Hawking radiation can generate a B−L asymmetry. Ghorbani & Ghorbani (2023) connect PBH formation with baryogenesis through an early matter-dominated era.

Crucial difference: All these models generate a *positive* baryon asymmetry (B > 0) through PBH processes. CBT postulates B = 0 as an axiom — the PBHs *hide* the negative side rather than generating the positive. This is a fundamental philosophical difference.

Zero-Energy-Universe

Tryon (1973) proposed that the universe might be a vacuum fluctuation with total energy zero. Guth (1981) argued in his inflation paper that negative gravitational energy compensates positive mass-energy. Hawking popularized the idea. CBT elevates this hypothesis to an axiom and links it for the first time systematically with PBH dark matter and B = 0.

What Makes the CBT Unique

The combination B = 0 + DM = PBHs + DE = zero sum has no direct predecessor in the published literature. The individual building blocks are known — the synthesis is new. In particular, the treatment of the baryon asymmetry as an observational artifact (rather than a phenomenon to be explained) and the resulting circumvention of the Sakharov conditions is a genuine conceptual contribution.

6. Open Tasks for Theoretical Physics

The CBT formulates a framework. The following tasks must be addressed by theoretical physics to fill the framework quantitatively:

1. Formation mechanism for PBHs in the asteroid-mass window. Which inflation model (or phase transition) produces PBHs at 10^{14}–10^{19} kg in the required abundance ($\Omega_{DM} = 0.265$)?

2. Correlation mechanism for B = 0. Which mechanism correlates baryon number fluctuations with density enhancements such that antimatter-excess regions preferentially collapse? Affleck-Dine is a candidate, not the only one.

3. Quantitative reproduction of $\Omega_{DM}/\Omega_b = 5.4$. The ratio must follow from the model parameters. This is the hardest falsification test.

4. Consistency with CMB, BBN, and gamma-ray limits. The PBH population must not violate observed limits (spectral distortions, light-element abundances, diffuse gamma-ray background).

7. Conclusion

The Cosmic Balance Theory reduces the three open categories of cosmology — dark matter, baryon asymmetry, dark energy — to two axioms. Mass is energy. The cosmos is a zero-sum game. Everything else follows from these or is an open calculation.

The theory is falsifiable: If PBHs in the asteroid-mass window are not found, it is refuted. If the net baryon number is demonstrably not zero, it is refuted. The instruments that can perform these tests exist or are being built.

> *The universe is not a puzzle with three unknowns. It is a zero-sum game with one rule: energy is conserved. What we see is the remainder after the collapse. What we do not see is the collapse itself.*

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